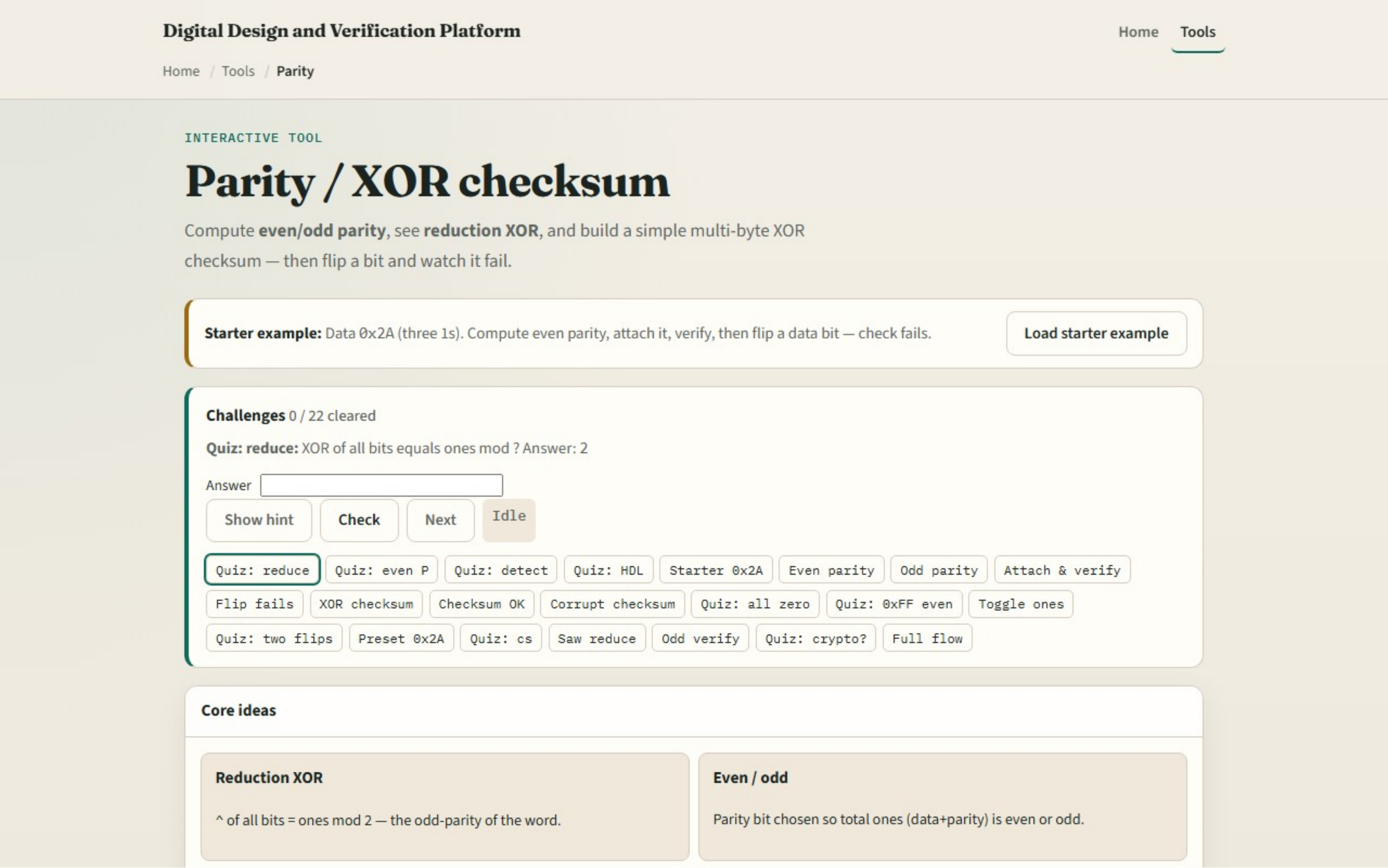
Parity and checksum

Links drop bits

Detect, do not correct

- Even parity sets the check bit so the total ones count is even
- Odd parity flips that sense
- XOR of all bits is ones-count mod two
- Parity does not correct bit errors
- A multi-byte XOR checksum verifies when the fold comes back zero

Browser lab



Parity checksum starter

Workbook practice

- In the workbook track, take data hex two-A
- Count the ones, compute even and odd parity bits by hand
- Sketch a three-byte XOR checksum and note that a corrupt byte fails the zero fold
- Name one pitfall: assuming parity repairs the bad bit

Pitfalls to watch

- Do not treat parity as error correction
- Even and odd modes disagree if you mix them across sender and receiver
- And remember: the browser lab is literacy
- Real links still need framing

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, compute one parity bit and one XOR checksum by hand
- When you are ready, take the short quiz, then continue to fixed-point